Design, Fabrication and analysis of Silicon based Imbalanced Mach Zehnder Interferometer and Microring resonator

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Abstract

This report is about designing, fabricating, optimizing and analyzing of Imbalanced Mach Zehnder Interferometer and Microring resonator on a silicon photonic integrated circuit. The device will be fabricated and tested by the program offered by SiEPIC-fab (Silicon Electronics Photonics Integrated Circuits fabrication).The measured values will be compared and analysed with the simulation results obtained using Lumerical MODE and INTERCONNECT.

1 Introduction

Silicon photonics is a technology platform that enables the integration of optical components on a silicon substrate using fabrication processes compatible with complementary metal oxide semiconductor (CMOS) technology. By exploiting the high refractive index contrast between silicon and silicon dioxide, silicon photonics provides strong optical confinement and enables compact device footprints. This compatibility with mature CMOS manufacturing processes allows scalable and cost-effective production of photonic integrated circuits (PICs), in which multiple optical functions like waveguides, couplers, modulators, filters and photodetectors are integrated on a single chip. Among the fundamental building blocks of silicon PICs is the Mach Zehnder interferometer (MZI), which consists of two optical splitters and two arms in which the relative phase of light can be controlled. In silicon photonics, MZIs are widely used for modulation, switching and filtering by inducing phase shifts through thermo-optic, carrier based effects or varying pathlength which precise control of optical interference at the output. In this work, we present the fabrication of Mach–Zehnder interferometers (MZIs) with different optical path lengths and analyze the resulting output variations as a function of phase changes that give rise to interference effects

2 Theory

An unbalanced Mach Zehnder interferometer (MZI) is a fundamental photonic device in which the two optical paths (arms) have unequal lengths, introducing a controlled phase difference between light propagating in each arm. This path length difference, ($\Delta L = L_2 - L_1$), produces wavelength-dependent interference at the output, which can be exploited for filtering, sensing, or wavelength-division multiplexing. Unlike balanced MZIs, which are primarily used as modulators or switches, unbalanced MZIs generate spectral interference fringes, enabling precise control over the transmitted intensity as a function of wavelength. The phase evolution of light in each arm is determined by the effective index (n_{eff}) of the guided mode, which depends on waveguide geometry and material properties. The phase difference between the two arms at a wavelength (λ) is expressed as:

$\Delta\phi = \frac{2\pi}{\lambda} n_{\text{eff}} \Delta L$

This phase difference directly affects the interference at the output: constructive interference occurs when $\Delta\phi = 2m\pi$ (with m an integer), resulting in maximum intensity, while destructive interference occurs when $\Delta\phi = (2m+1)\pi$, producing intensity minima.

The group index ((n_g)) defines the propagation of optical pulses or modulated signals in the waveguide,

accounting for material and waveguide dispersion:

$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}$.

The group index also determines the free spectral range (FSR)of the unbalanced MZI, which is the wavelength spacing between adjacent intensity maxima:

$\text{FSR} = \frac{\lambda^2}{n_g \Delta L}$.

A larger path-length difference (ΔL) results in a smaller FSR, producing closely spaced interference fringes in the transmission spectrum.

The intensity-based transfer function of an ideal lossless unbalanced MZI is given by:

$I_{\text{out}}(\lambda) = I_{\text{in}} \cos^2\left(\frac{\pi n_{\text{eff}} \Delta L}{\lambda}\right)$,

where I_{in} is the input intensity and I_{out} is the output intensity at a given wavelength. By carefully designing ΔL, n_{eff}, and n_g, unbalanced MZIs can be tailored to achieve desired spectral characteristics, making them essential components in silicon photonic circuits for filtering, sensing, and wavelength-selective routing

A microring resonator is wavelength-selective cavitiy that enhance light–matter interaction. It consist of closed-loop waveguide, traps light by continuous total internal reflection and supports resonant circulation at specific wavelengths. When a straight waveguide is placed close to the ring, light can couple evanescently into it, but only wavelengths that satisfy the resonance condition build up constructively. These resonant wavelengths depend on the ring's radius, effective refractive index, and losses and they show up as sharp transmission dips or peaks in the coupled waveguide spectrum. Small changes in refractive index by temperature, electric fields or the presence of analytes can shift the resonance wavelengths, which makes microrings powerful for modulation, switching and sensing. Their compatibility on photonic chips and small footprint enables applications such as filters, wavelength-division multiplexing, compact lasers, nonlinear optics and biosensors. For a microring resonator, the resonance condition is obtained when accumulated phase shift equals an integer multiple of 2π.

$m\lambda = 2\pi R n_{eff}$

m is the mode number, R is the radius of the ring and n_{eff} is the effective refractive index.

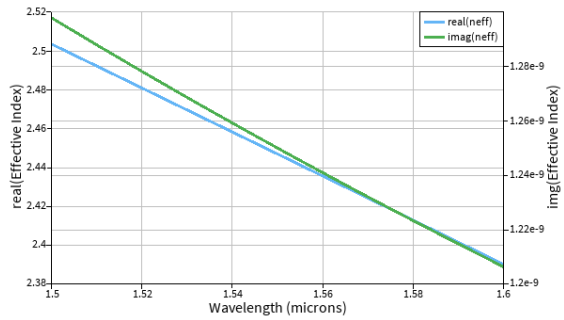

Figure 1: Effective index simulated using Lumerical MODE

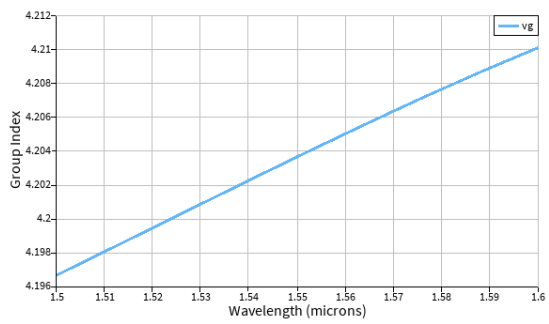

Figure 2: Group index simulated using Lumerical MODE

3 Modelling and Simulation

The waveguide dimensions are width=500nm and thickness=220nm with TE polarization. The operating wavelength is 1550nm.

The fitted model for effective index in wzaveguide compact model is obtained as:

$2.444 - 1.130(\lambda - 1.55) - 0.038(\lambda - 1.55)^2$

The effective index and group index dependency on wavelength for silicon waveguide is plotted using Lumerical MODE. The layout for imbalanced MZI configurations is constructed in Klayout and corresponding spectrum is obtained in Lumerical INTERCONNECT.

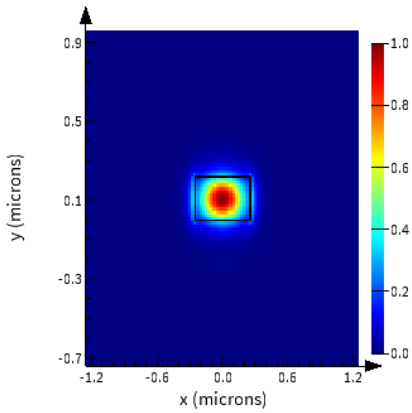

Figure 3: Mode profile for first order TE mode

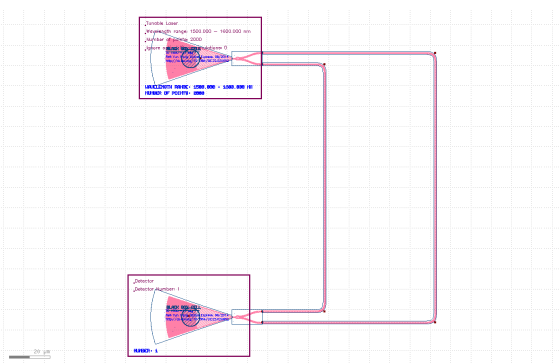

Figure 4: Simple MZI layout in Klayout

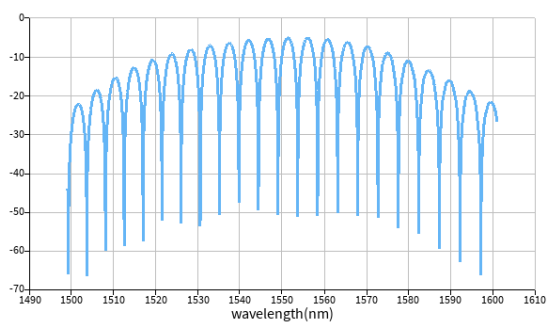

Figure 5: Output spectrum at the detector using Lumerical INTERCONNECT for ΔL =120nm

Table 1: MZI length variation and FSR at 1550nm

ΔL (μm)	FSR(nm)
270	2.120
330	1.742
422	1.361

4 Fabrication

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a Raith EBPG 5000+ electron beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4” handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate

(TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

5 Simulation Data

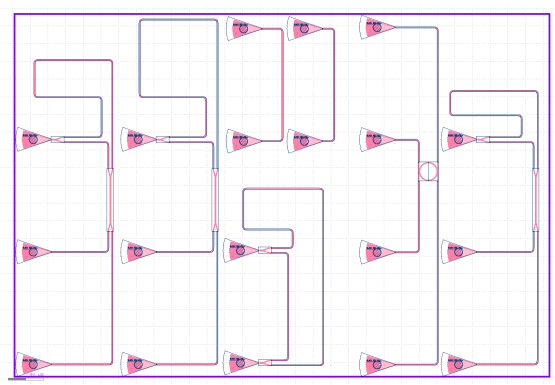

Figure 6: layout in Klayout

The layout consists of 2 direct connected grating couplers, 4 set of MZI and a ring resonator of radius 10μm. The first MZI in the left is named as MZI1 followed by MZI2 and then the waveguides directly connected. The MZI below the waveguide is named as MZI3.

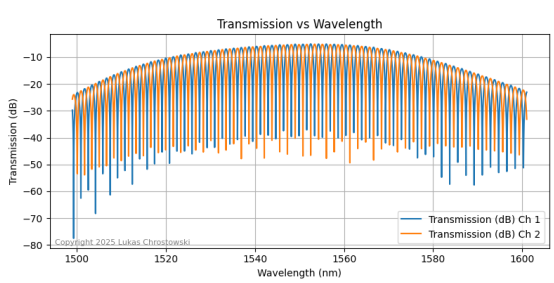

Figure 7: output spectrum of MZI1

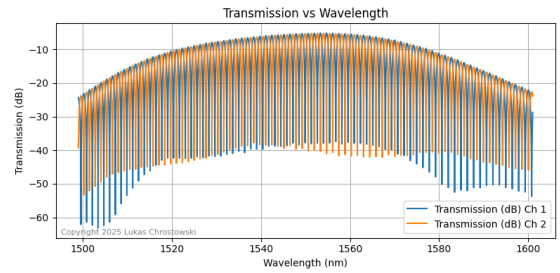

Figure 8: output spectrum of MZI2

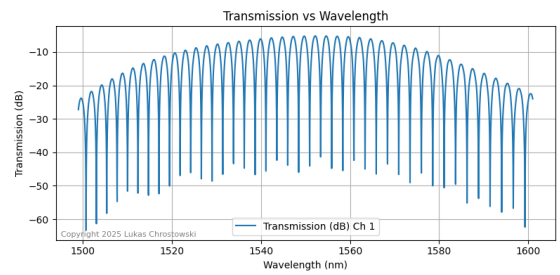

Figure 9: output spectrum of MZI3

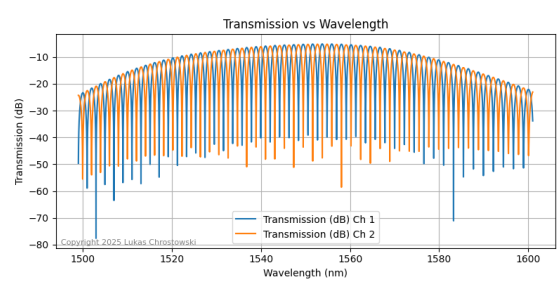

Figure 10: output spectrum of MZI6

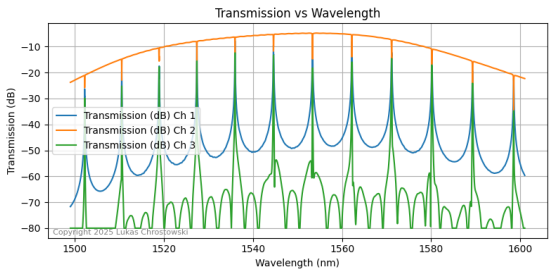

Figure 11: output spectrum of Ring resonator

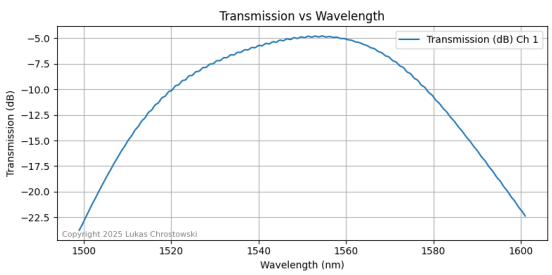

Figure 12: output spectrum of waveguide1

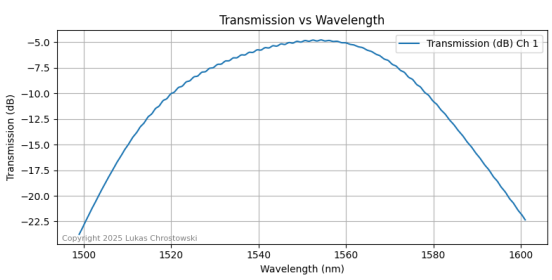

Figure 13: output spectrum of waveguide2

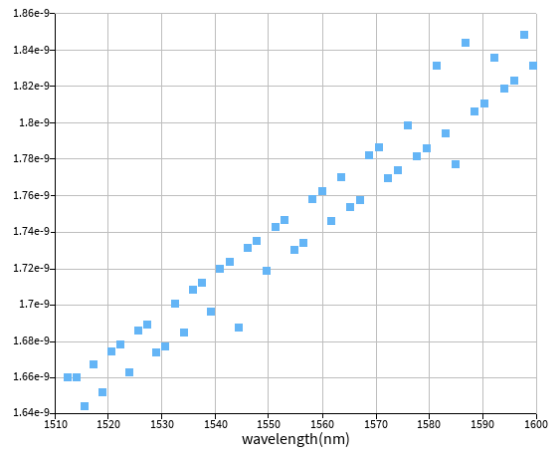

Figure 14: FSR of MZI1

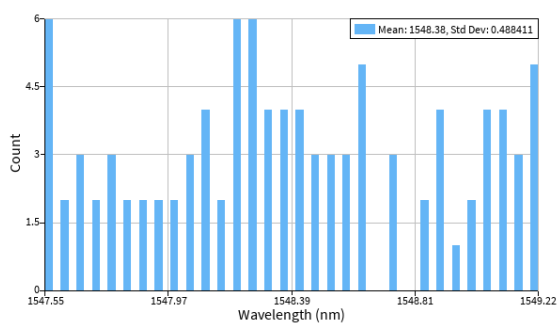

Figure 15: Monte carlo simulation

6 Experimental Data and Analysis

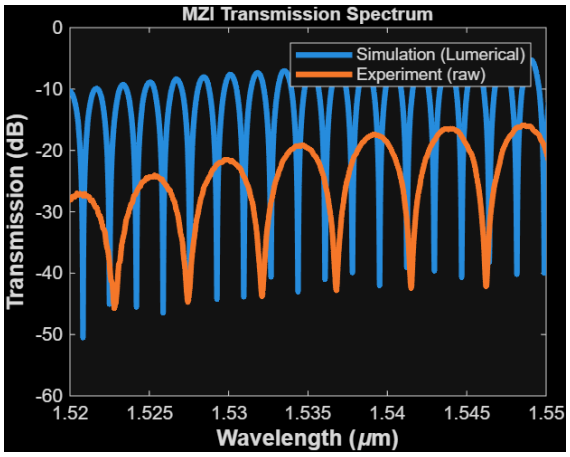

Figure 16: comparison of experimental and simulation results in MZI1

The experimental results doesnot match with the simulation results for the MZI1 in the klayout. The FSR in experimental results seems higher than the simulation results.

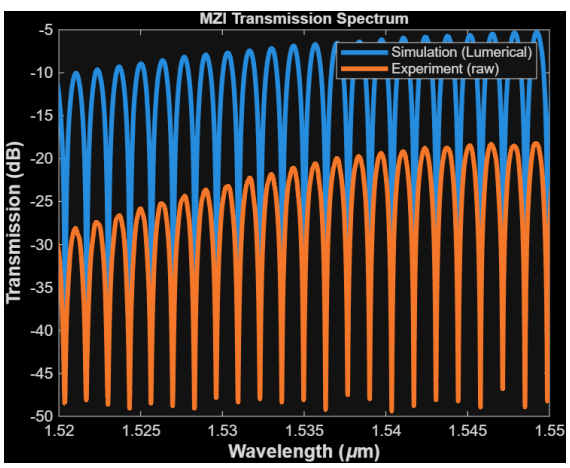

Figure 17: comparison of experimental and simulation results in MZI2

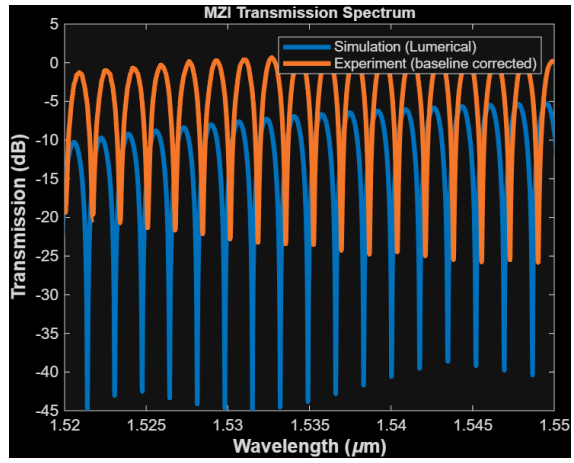

Figure 18: comparison of experimental and simulation results in MZI3

The experimental and simulation results for MZI2 and MZI3 are aligning with only a small shift with in the tolerance limit.

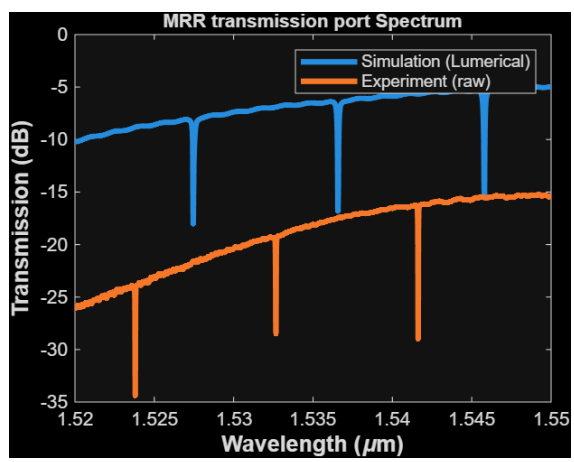

Figure 19: comparison of experimental and simulation results in Tport of Ring resonator

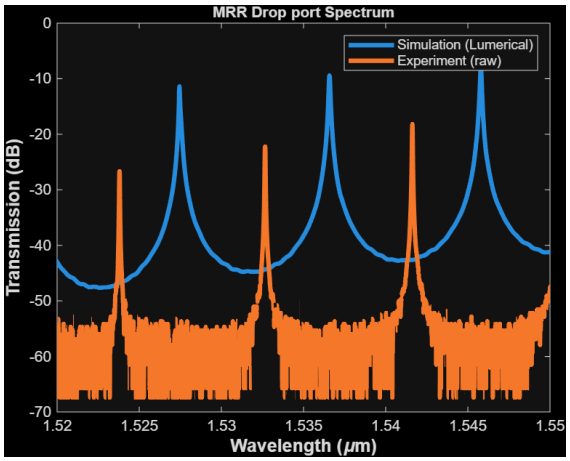

Figure 20: comparison of experimental and simulation results in Dport of Ring resonator

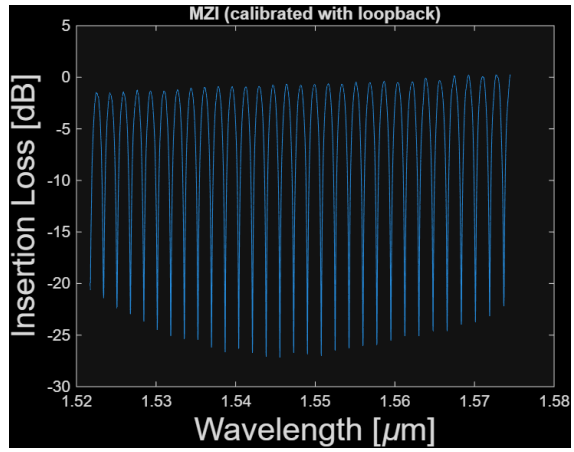

Figure 22: Calibrated loopback output in MZI3

The experimental and simulation results for microring resonators are not quite matching. The resonant frequencies are different.

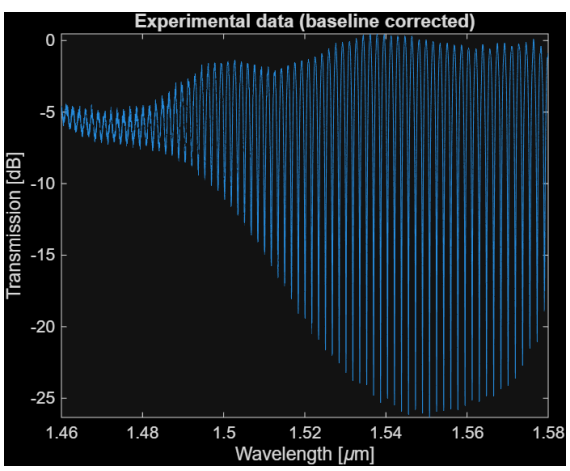

Figure 21: Baseline corrected output in MZI3

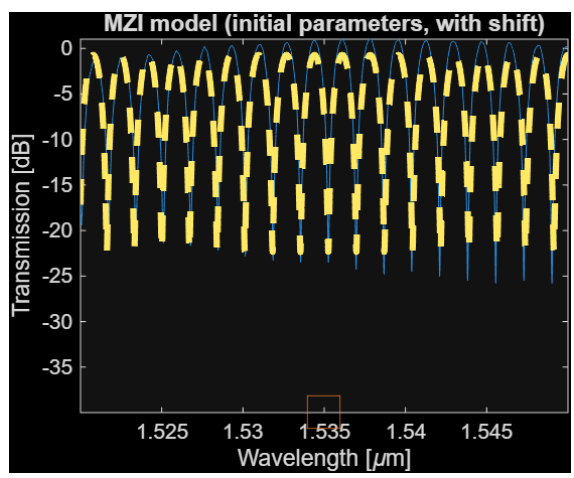

Figure 23: MZI3 correlation initial with shift

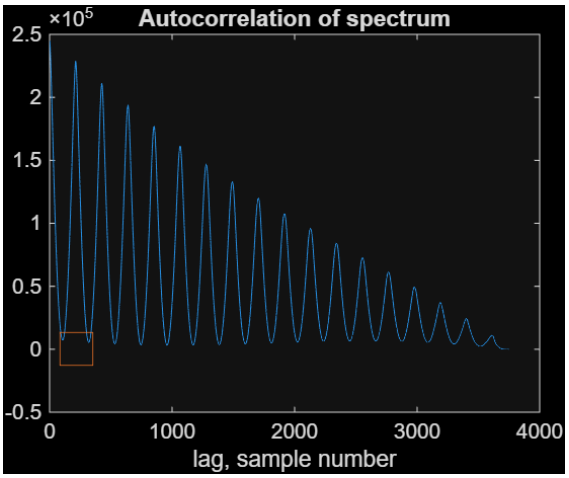

Figure 24: Autocorrelation spectrum for MZI3

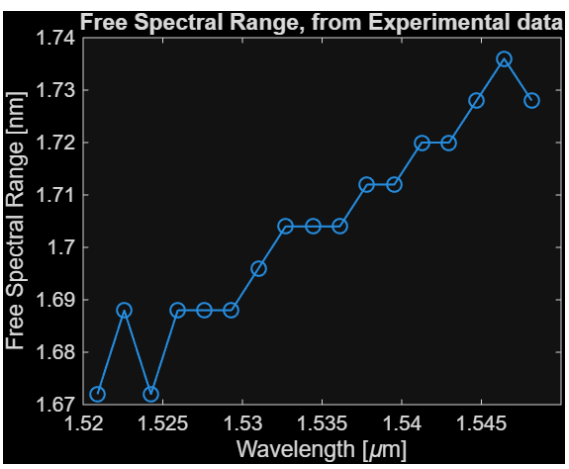

Figure 25: FSR of MZI3 using experimental data

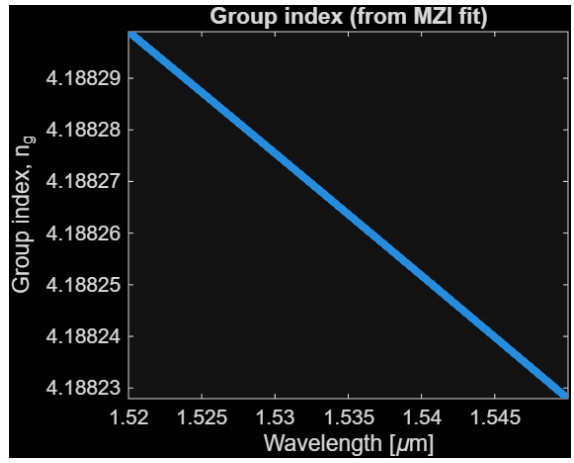

Figure 26: Group index of MZI3 using experimental data

The n1,n2 and n3 obtained for the polynomial fit from the experimental data are 2.454,-1.129 and 0.0008. The obtained experimental result based FSR for MzI3 is 1.72nm while the simulation based FSR us 1.74nm. The obtained experimental result based FSR for MzI2 is 1.360nm while the simulation based FSR us 1.361nm. The group index obtained is 4.12. The experimental and simulation results for microring resonator of radius 10μm is having a mismatch of 3nm shift.

7 Conclusion

The experimental outputs of MZI2 and MZI3 are matching with the simulation results and the obtained FSR and group index are highly matching. The experimental results for MZI1 is having significant variation from the expected results. The experimental results for microring resonator is showing resonant frequencies which are a shifted version of the simulated resonant frequencies.

References

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